Pediatric Surgery International (2024) 40:120
https://doi.org/10.1007/s00383-024-05694-0

ORIGINAL ARTICLE

Is neurodevelopment impaired in Brazilian children with intestinal failure on prolonged parenteral nutrition? A single center study

Cristina Miller[1] · Renata R. Kieling[2] · Bruna Ziegler[1] · Juliana M. Giesta[3] · Alana V. Signorini[2] · Giovana Q. Pires[2] · Letícia Feldens[1] · Marília R. Ceza[1] · Marina R. Adami[1] · Carlos O. Kieling[1] · Helena A. S. Goldani[1]

Accepted: 8 April 2024 / Published online: 4 May 2024

Abstract

Purpose To assess the neurodevelopment outcomes of children younger than 42 months of age with intestinal failure (IF) using prolonged parenteral nutrition (PN) followed by a Pediatric Multidisciplinary Intestinal Rehabilitation Program from a public tertiary hospital in Brazil.

Methods Bayley III scale was administered in children aged 2 to 42 months with IF and receiving PN for more than 60 days. Composite scores in cognitive, motor, and language domains were analyzed. Developmental delay was defined as a performance 2 standard deviations (SD) below the average at the 3 domains. Association between Bayley III composite scores and clinical variables related to IF were tested.

Results Twenty-four children with median (IQR) age of 17.5 months (9–28.5) were studied, 58.3% were male. Developmental delay was found in 34%, 33% and 27% of the patients in cognitive, motor, and language domains, respectively. There was no significant association between the Bayley-III composite scores and length of hospitalization, prematurity, and number of surgical procedures with anesthesia.

Conclusion The study demonstrated impairments in the cognitive, motor and language domains in approximately one-third of young patients with IF on prolonged PN.

Keywords Neurodevelopment · Intestinal failure · Child · Parenteral nutrition · Bayley-III

Introduction

Intestinal Failure (IF) is a complex and serious medical condition that occurs when the gastrointestinal tract is unable to absorb nutrients and fluids necessary for sustaining life, resulting in dependence on parenteral support for a minimum of 60 days [1–3]. IF is considered a rare condition, with worldwide prevalences ranging from 14.1 to 56 cases of pediatric chronic IF per million children [1, 2, 4, 5]. IF can result from various underlying causes, including congenital conditions, surgical resections, or functional disorders. Short bowel syndrome (SBS) is the most frequent cause of IF, with the most common etiologies being necrotizing enterocolitis, intestinal atresia, volvulus, and gastroschisis [6, 7].

The clinical management of IF involves the use of PN, medications to control symptoms and fluid balance, and enteral nutrition to promote intestinal adaptation. Surgical interventions can improve intestinal adaptation. Patients who do not respond to this management, those with limited potential for intestinal adaptation, or those who develop intractable complications become potential candidates for intestinal and multivisceral transplantation. Recent data from European and North American intestinal rehabilitation centers have shown an increase in the long-term survival rate to more than 90% [1, 5, 7].

There is a scarcity of studies evaluating the neurodevelopment of children with IF in use of prolonged PN, particularly in low resource settings from middle-income countries. Several studies in children with IF have suggested the presence of neurodevelopmental delays, both during the first year of life and in the school-age period. Additionally, these studies

✉ Cristina Miller
cmiller@hcpa.edu.br

1 Hospital de Clínicas de Porto Alegre, Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil

2 Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil

3 Hospital de Clínicas de Porto Alegre, Porto Alegre, Brazil

indicated that several clinical variables associated with pediatric IF might influence motor skills and physical activity at school [8–10].

The aim of this study was to assess the neurodevelopment of a cohort of Brazilian children with IF using prolonged PN aged less than 42 months from a public tertiary hospital in Brazil. Additionally, the study aimed to identify clinical factors that might be related to the neurodevelopmental profile of these children.

Methods

This is a contemporary cross-sectional study of IF patients using prolonged PN. The study was approved by the Research Ethics Committee of the Hospital de Clínicas de Porto Alegre, CAAE no. 43916621700005327. Informed and written consent was obtained from the parents or legal guardians of the children.

Study population

The participants are part of a cohort of patients followed at the Intestinal Rehabilitation Program of Children and Adolescents at Hospital de Clínicas de Porto Alegre (PRICA-HCPA). For this study, children were evaluated at a single time point either during their hospitalization in the Pediatric Hospital Unit or during routine outpatient consultations.

Between June 2021 and October 2022, we selected patients followed by PRICA-HCPA who met the following inclusion criteria: children aged 2 to 42 months with a diagnosis of IF and receiving prolonged PN for more than 60 days. Children diagnosed with severe neurological disease, severe congenital malformations, genetic and metabolic syndromes with previous neurodevelopmental impairment were excluded.

Eligible patients were invited to participate in the study, and after obtaining informed consent, the evaluation was conducted following the guidelines for assessing the cognitive, motor, and language domains of the Bayley Scales of Infant and Toddler Development, Third Edition (Bayley-III) [11–13].

Clinical data

Clinical and demographic data were obtained from electronic medical records. The characteristics of the patients included in the study were: age; sex; birth weight; gestational age at birth; nutritional status at the time of Bayley-III assessment; length of hospital stay; and duration of PN use. Clinical data related to IF included: cause of IF; length of the remaining intestine; number of central venous catheter (CVC) replacements; comorbidities; number of sepsis episodes; and number of hospital admissions.

Bayley-III scale

The Bayley-III scale assesses various domains of neurodevelopment and is based on a robust theoretical framework with strong psychometric properties. The normative data for the Bayley-III were collected in the US in 2004 [12], and a trans cultural adaptation for the use of the third version in Brazil was carried out in 2016 [14].

The scale is administered individually and assesses the developmental functioning of infants and young children aged between 1 month and 42 months. The primary objective is to identify children with developmental delays and provide support for intervention planning [12]. The Bayley-III scale assesses cognitive, motor (fine motor and gross motor subtests) and language (receptive and expressive language subtests) domains [12, 13].

The composite score enables the distribution of the scores among performance level categories, based on mean and standard deviation (SD) for each domain. This score is calculated based on a comparison of the child to a normative age-matched sample and can be interpreted according to the criteria presented by Del Rosario et al [11] as follows: a) 100 points (SD=15), corresponding to the standard mean score (50th percentile), indicating functioning within the average range; b) below 85 points, corresponding to 1 SD below the average (16th percentile), indicating mild impairment or being 'at risk' of developmental delay requiring follow-up, parental counseling on techniques to enhance development, and referral to therapists depending on the degree of impairment; c) below 70 points, corresponding to 2 SD below the mean (2nd percentile), indicating moderate/severe impairment requiring early intervention by the healthcare team. The results of the composite scores are derived from the sum of several scores of subtest scales. They were generated for the Cognitive, Motor and Language domains, ranging between 40 and 160 [12].

In this study, the Bayley-III scale was administered by the same investigator (CM), a physiotherapist with prior training and certification in the use of Bayley-III scale. The assessments were performed at a single time point, either in an outpatient setting or in a controlled hospital environment, with minimal environmental stimulants or stressors that could potentially affect the test results. Those children evaluated in the hospital setting were assessed close to their hospital discharge to ensure they were in the best physical condition for the assessment. All evaluations took place in accordance with the scale's application standards.

Statistical analysis

The characteristics of the population were analyzed as follows: continuous quantitative variables were described using the median (IQR), while variables with a normal distribution were presented as mean (SD). Categorical variables were described using percentages and frequencies. We evaluated the performance results of the Bayley-III scale using composite scores, presented as mean (SD). For a more specific analysis, a z-score based on the composite scores was calculated for each domain. To examine the correlation between the results of the Bayley-III scale and clinical factors related to neurodevelopmental outcomes, we employed Spearman's rho correlation test, Student's t-test, and ANOVA (analysis of variance), when appropriate. Statistical analyses were conducted using SPSS software version 29.0, and the significance level was set at $P \leq 0.05$.

Results

Twenty-four children with IF on prolonged PN were evaluated. The median (IQR) age of the participants at the time of evaluation was 360.5 days (274.7–871) or 17.5 (9–28.5) months. Most of the patients were male (58.3%), and the median (IQR) gestational age was 35.5 (32,2–38) weeks. A total of 58.3% of the patients underwent evaluation on an outpatient basis. The clinical characteristics of the sample are presented in Table 1.

The cognitive subtest was applied to 95.8% (n=23) of our population, the motor subtest was administered to 100% of our sample and the language subtest was applied to 91.7% (n=22) children in the sample.

Regarding the valid results of the cognitive performance, it was observed that 47.8% of the children presented average performance, 17.4% presented 1 SD below the mean, indicating a risk of developmental delay, and 34.8% presented performance 2 SD below the mean, indicating moderate to severe delay. At the motor performance subtest, the results indicated that 45.8% of the children demonstrated average performance, 20.8% 1 SD below the mean, indicating a risk of delay, and 33.3% had performance 2 SD below the mean, indicating a risk of moderate to severe delay. Finally, regarding the language performance valid results, it was found that 50.0% of the children presented average performance, 22.7% demonstrated 1 SD below the mean, indicating a risk of delay, and 27.3% presented performance 2 SD below the mean, indicating a risk of moderate/severe delay (Table 2).

The results of the associations between some clinical variables and the results of Bayley-III composite scores in the three domains were presented in Table 3. There was no significant association among the results of performances in the cognitive, motor and language domains total and the

Table 1 Clinical characteristics of patients

Descriptive features	
Male n (%)	14 (58.3)
Median (IQR) age at the time of assessment (days)	360.5 (274.7–871)
Age categories at the time of assessment—n (%)	
< 12 months	13 (54.2)
12 to < 24 months	3 (12.5)
24 to < 36 months	6 (25.0)
36 to 42 months	2 (8.3)
Mean (SD) time of use of PN (days)	415 (269.2–756)
Median (IQR) total length of hospital stay (days)	299 (265.2)
Gestational age al birth n (%)	
< 28 weeks—extremely preterm	2 (8.3)
28 to < 32 weeks—very preterm	3 (12.5)
32 to < 37—moderate to late preterm	11 (45.8)
≥ 37 weeks—term	8 (33.3)
Birth weight—n (%)	
< 1.5 kg	2 (8.3)
1.5 to < 2.5 kg	9 (37.5)
≥ 2.5 kg	13 (54.2)
Weight for gestational age at birth—n (%)	
SGA	4 (16.7)
AGA	18 (75)
LGA	2 (8.3)
Nutritional status in the assessment—n (%)	
Eutrophic	17 (70.8)
Undernourished	7 (29.2)
Cause of IF—n (%)	
Gastroschisis	9 (37.5)
Volvo	6 (25)
Necrotizing enterocolitis	3 (12.5)
Atresia	3 (12.5)
Hirschsprung	1 (4.2)
Meconium ileus	1 (4.2)
Other	1 (4.2)
Remaining intestine—n (%)	
< 40 cm	9 (37.5)
≥ 40 cm	15 (62.5)
No. of CVC replacements—n (%)	
< 4	16 (66.7)
≥ 4	8 (33.3)
Comorbidities—n (%)	
Pulmonology	4 (16.7)
Hepatic impairment	1 (4.2)
Hearing impairment	1 (4.2)
Visual alteration	3 (12.5)
Neurological	7 (29.2)
Sepsis—n (%)	
Yes	17 (70.8)
No. of hospital admissions—n (%)	
< 4	20 (83.3)

Table 1 (continued)

Descriptive features	
≥4	4 (16.7)
No. of surgeries—n (%)	
<4	19 (79.8)
≥4	5 (20.8)

SGA small for gestational age, *AGA* adequate for gestational age, *LGA* large for gestational age, *IF* intestinal failure, *CVC* central venous catheter

Table 2 Performance results of Bayley-III composite scores in the three domains

Performance	No impairment[a] n (%)	Mild impairment[b] n (%)	Moderate/severe Impairment[c] n (%)
Cognitive	11 (47.8)	4 (17.4)	8 (34.8)
Motor	11 (45.8)	5 (20.8)	9 (33.3)
Language	11 (50.0)	5 (22.7)	6 (27.3)

[a]Within the average range

[b]1SD below the mean

[c]2 SD below the mean

Table 3 Results of the associations between clinical variables and the mean composite scores of Bayley–III

Clinical variables	Cognitive	Motor	Language
Total length of hospital stay[a]	−0.168	−0.251	−0.241
P	0.443	0.261	0.256
Time of PN use[a]	−0.231	−0.313	−0.097
P	0.289	0.157	0.651
Gestational age at birth[c]			
<28 weeks	−2.16 ± 1.7	−3.20 ± 0.5	−2.26 ± 1.2
28 to <32 weeks	−2.33 ± 0.6	−1.86 ± 1.6	−2.40 ± 0.1
32 to <37 weeks	−0.80 ± 1.0	−1.04 ± 1.1	−0.81 ± 0.7
37 weeks	−0.95 ± 1.2	−1.60 ± 1.3	1.07 ± 1.2
P	0.138	0.193	0.069
No. of surgeries[b]			
<4	−1.26 ± 1.34	−1.44 ± 1.35	−1.55 ± 1.7
≥4	1.04 ± 0.9	−1.63 ± 1.37	−0.83 ± 0.8
P	0.196	0.691	0.144
No. of CVC replacements[b]			
<4	−1.13 ± 0.9	−1.6 ± 1.4	−1.2 ± 1.0
≥4	−1.21 ± 1.5	−1.45 ± 1.4	−1.34 ± 1.2
P	0.887	0.821	0.758

[a]Spearman's *rho*

[b]Student's *t*-test

[c]Anova

following clinical variables: length of hospital stay, time of PN use, gestational age at birth, number of surgeries, and number of CVC replacements.

Discussion

This study used the Bayley-III scale to assess several aspects of neurodevelopment of children with IF receiving prolonged PN followed by a reference center for intestinal rehabilitation in Brazil, a pioneering initiative in the Brazilian public health system. The main finding showed that approximately one-third of the children presented moderate/severe development performance in cognitive, motor, and language domains, indicating a functional impairment characterized by moderate/severe delay, requiring ongoing monitoring.

Previous studies that assessed the neurodevelopment of children with IF, employing various assessment instruments, showed a prevalence of impairment in neuromotor development ranging from 37% to 60% [2, 8, 9]. Our study yielded a similar outcome, using the Bayley III scale, a comprehensive developmental assessment scale that provides insights into cognitive, motor, and language performance.

In 2016, Chesley *et al* presented the first data regarding the neurodevelopment of children with IF. They evaluated 15 children aged between 12 and 67 months and found that 80% of the children had cognition and neurodevelopment within the normal range, with scores 1SD or 2SD below the mean. However, taking into account that 20% of those assessed had severe neurological comorbidities, a reanalysis of the data might demonstrate that approximately 60% of the sample would have scores 1SD or 2SD below the normative average [15–17].

In our study, we found that 47.8% of the children had mean cognitive performance within the normal range. However, 52.2% showed performance 1SD or 2SD below the mean, indicating a mild risk of delay or moderate to severe neurodevelopmental impairment. Our findings are consistent with those reported by Gold et al, where 46% of the sample showed cognitive alterations in specific tests [15].

Motor performance was also evaluated by So *et al* in 33 children with IF aged between 4 and 15 months, using three specific scales. Among the children assessed with the Prechtl's Assessment of General Movements scale, 88% exhibited abnormal movements, which could be indicative of long-term cognitive and motor dysfunction. In those evaluated using the Alberta Infant Motor scale, 53% presented altered scores, indicating delayed psychomotor development. Furthermore, the results from the Mullen Assessment scale indicated significant delays in the motor scale, with 50% of the participants presenting significant delay and 31% exhibiting moderate delay [8].

The outcomes of our current study are in line with those reported by So *et al* as we found that 45.8% of the children presented average functional motor performance, 20.8% showed performance 1SD below the mean, suggesting a risk of delay, and 33.3% showed performance 2SD below the mean, indicative of moderate/severe impairment [8].

In 2019, So *et al.* found that those developmental delays may persist over time as the authors reassessed 31 children at 12 and 26 months of age using the Mullen Assessment Scale and the Vineland Scale II. Regarding the motor domain, 51.9% scored below the average at 26 months [9]. The same authors found that 50% of school-age children had below-average scores in motor skills [10]. Therefore, those studies emphasized the importance of assessing the risk of neurodevelopmental delays in children with IF receiving prolonged PN.

Regarding the performance in the language domain, we found that 50% of the children demonstrated average performance, 22.7% had performance with mild impairment or a risk of delay, and 27.3% had performance with moderate/severe impairment. Notably, no other study has provided specific data regarding expressive and receptive communication in children with IF on prolonged PN.

Several factors have been hypothesized to be responsible for the association between neurodevelopmental delays and IF. The process of learning through experiences contributes to the ability to assimilate specific information and formulate appropriate responses required for living in different environments [18, 19]. As child development is a multifactorial process, changes in the environment and physiological alterations can impact neural connectivity and cognitive processing in developing systems. Disruptions during early childhood can have long-term effects on development, affecting both the structural and functional capacity of the central nervous system, thereby increasing the risk of neurodevelopmental delays [20].

In 2020, Bell *et al* found that infants with SBS born at 22–26 weeks of gestational age had lower cognitive, language, and motor scores compared to children with other bowel diseases without SBS. Additionally, they demonstrated that premature infants with SBS were more likely to experience neonatal morbidities associated with neurodevelopmental delays [21]

Children following surgical procedures are at risk of neurodevelopmental delay as exposure to anesthesia, postoperative systemic inflammation and inadequate pain control have been linked to changes in the early development of the central nervous system. One of the complications in children with IF on prolonged parenteral nutrition is the recurrent catheter-related bloodstream infections, which can lead to recurrent exposure of these children to periods of anesthesia and systemic inflammation [22, 23]

Several factors were considered as potential negative influences on the neurodevelopment of children with IF, including prematurity, the underlying disease and age of symptom onset, practices related to prolonged PN administration, age at the initiation of PN, comorbidities, extended hospitalization periods, as well as multiple surgeries and procedures [15, 16]. In our study, we did not find any significant association between the observed neurodevelopmental alterations and specific clinical conditions of children with IF, including gestational age at birth; duration of hospitalization and duration of PN use; number of surgeries; frequency of central venous catheter (CVC) replacements, and initiation of PN before 60 days of life.

This lack of association may be attributed to the small sample size of our study, which is the main limitation of this study. Multicenter studies may be needed to better evaluate it; nevertheless, the strength of this study lies in its ability to corroborate the high occurrence of neurodevelopmental delays in children with IF who are on prolonged PN use, particularly among patients in the context of the public health system in a middle-income country.

In conclusion, this study demonstrated impairments in the cognitive, motor, and language performances in approximately one-third of children with IF using prolonged PN in a reference center for intestinal rehabilitation in Brazil. These results are similar to findings from intestinal rehabilitation centers worldwide and address the imperative need for the evaluation and ongoing monitoring of the neurodevelopment of these patients.

Acknowledgements We are grateful to all participating children and their families for dedicating their valuable time to this study. We also appreciate the valuable contribution of the multiprofessional team from the local Program of Intestinal Rehabilitation.

Author contributions CM, RRK, HASG participated in the conception and design of the study. CM, BZ, JMG, AVS, GQP, LF, MRC, MRA, COK participated in the data collection. CM, RRK, COK, HASG participated in analysis and interpretation of data. CM and HASG produced the first draft of the manuscript and all authors were involved in producing the final version and approved its content.

Funding This study was supported by Fundo Nacional da Saúde, Ministry of Health; Coordenação de Aperfeiçoamento de Pessoal de Nível Superior—Brasil (CAPES) and FIPE—Hospital de Clinicas de Porto Alegre.

Data availability Requests for data sharing will be considered upon written request to the corresponding author.

Declarations

Conflict of interest The authors declare no conflict of interest.

References

1. Belza C, Wales PW (2020) Multidisciplinary management in pediatric ultrashort bowel syndrome. J Multidiscip Healthc 13:9–17. https://doi.org/10.2147/JMDH.S236130

2. Goldani HAS, Ceza MR, Godoy LL et al (2022) Outcomes of the first 54 pediatric patients on long-term home parenteral nutrition from a single Brazilian center. J Pediatr Gastroenterol Nutr 75:104–109. https://doi.org/10.1097/MPG.0000000000003473
3. Modi BP, Galloway DP, Gura K et al (2022) ASPEN definitions in pediatric intestinal failure. JPEN J Parenter Enteral Nutr 46:42–59. https://doi.org/10.1002/jpen.2232
4. Lezo A, Diamanti A, Marinier EM et al (2022) Chronic intestinal failure in children: an international multicenter cross-sectional survey. Nutrients. https://doi.org/10.3390/nu14091889
5. Fullerton BS, Sparks EA, Hall AM et al (2016) Enteral autonomy, cirrhosis, and long term transplant-free survival in pediatric intestinal failure patients. J Pediatr Surg 51:96–100. https://doi.org/10.1016/j.jpedsurg.2015.10.027
6. Duggan CP, Jaksic T (2017) Pediatric intestinal failure. N Engl J Med 377:666–675. https://doi.org/10.1056/NEJMra1602650
7. Mangalat N, Teckman J (2018) Pediatric intestinal failure review. Children 5:100. https://doi.org/10.3390/children5070100
8. So S, Patterson C, Gold A et al (2016) Early neurodevelopmental outcomes of infants with intestinal failure. Early Hum Dev 101:11–16. https://doi.org/10.1016/j.earlhumdev.2016.05.012
9. So S, Patterson C, Gold A et al (2019) Neurodevelopmental outcomes of infants with intestinal failure at 12 and 26 months corrected age. Early Hum Dev 130:38–43. https://doi.org/10.1016/j.earlhumdev.2018.12.020
10. So S, Patterson C, Evans C, Wales PW (2019) Motor proficiency and generalized self-efficacy toward physical activity in children with intestinal failure. J Pediatr Gastroenterol Nutr 68:7–12. https://doi.org/10.1097/MPG.0000000000002107
11. Del Rosario C, Slevin M, Molloy EJ et al (2021) How to use the Bayley scales of infant and toddler development. Arch Dis Child Educ Pract Ed 106:108–112. https://doi.org/10.1136/archdischild-2020-319063
12. Bayley N (2018) Bayley scales of infant and toddler development Technical manual, 3rd edn. Pearson Clinical Brasil, São Paulo
13. Bayley N (2018) Bayley scales of infant and toddler development Administration manual, 3rd edn. Pearson Clinical Brasil, São Paulo
14. Madaschi V, Mecca TP, Macedo EC, Paula CS (2016) Bayley-III scales of infant and toddler development: transcultural adaptation and psychometric properties. Paideia 26:189–197. https://doi.org/10.1590/1982-43272664201606
15. Gold A, Danguecan A, Belza C et al (2020) Neurocognitive functioning in early school-age children with intestinal failure. J Pediatr Gastroenterol Nutr 70:225–231. https://doi.org/10.1097/MPG.0000000000002500
16. Vlug LE, Verloop MW, Dierckx B et al (2022) Cognitive outcomes in children with conditions affecting the small intestine: a systematic review and meta-analysis. J Pediatr Gastroenterol Nutr 74:368–376. https://doi.org/10.1097/MPG.0000000000003368
17. Chesley PM, Sanchez SE, Melzer L et al (2016) Neurodevelopmental and cognitive outcomes in children with intestinal failure. J Pediatr Gastroenterol Nutr 63:41–45. https://doi.org/10.1097/MPG.0000000000001067
18. Fox SE, Levitt P, Nelson CA (2010) How the timing and quality of early experiences influence the development of brain architecture. Child Dev. 81:28–40. https://doi.org/10.1111/j.1467-8624.2009.01380.x
19. Mangin KS, Horwood LJ, Woodward LJ (2017) Cognitive development trajectories of very preterm and typically developing children. Child Dev 88:282–298. https://doi.org/10.1111/cdev.12585
20. Maggi EF, Magalhães LC, Campos AF, Bouzada MCF (2014) Preterm children have unfavorable motor, cognitive, and functional performance when compared to term children of preschool age. J Pediatr (Rio J) 90:377–383. https://doi.org/10.1016/j.jped.2013.10.005
21. Bell M, Cole CR, Hansen NI et al (2020) Neurodevelopmental and growth outcomes of extremely preterm infants with short bowel syndrome. J Pediatr 230:76-83.e5. https://doi.org/10.1016/j.jpeds.2020.11.026
22. Lo E, Kalish BT (2023) Neurodevelopmental outcomes after neonatal surgery. Pediatr Surg Int. https://doi.org/10.1007/s00383-022-05285-x
23. Sakurai T, Nakamura M, Sasaki H et al (2023) Risk factors for catheter-related bloodstream infections in patients with intestinal failure undergoing home parenteral nutrition: a single-center study. Pediatr Surg Int. https://doi.org/10.1007/s00383-023-05555-2

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.